Appendix B. Evidence Engine, Recipe Ranking and Scenarios (Layers 3–5)

This appendix documents how the HepatoTwin pipeline turns the Layer 2 patient state into evidence-linked nutritional priorities (Layer 3), ranks candidate recipes from RecipeDB against them (Layer 4, NutriTwin), builds seven-day meal plans, and compares recipe choices as scenarios (Layer 5). It records the evidence base, the rules, the recipe criteria and the screening of the recipe corpus, and reports the evaluation set out in Section 12.2 of the proposal. Like the Layer 2 model (Appendix A), the pipeline is evaluated on held-out patients aged 12 to 59, the model's scope. All figures come from scripts and result files in the project repository.

B.1 Pipeline Overview

1. **Layer 2, patient state.** The combined model (Appendix A) gives a predicted CAP and calibrated probabilities P(CAP ≥ 248) and P(CAP ≥ 280), each with an F1-selected decision threshold.
2. **Layer 3, patient-state flags.** The probabilities and routine measurements are converted into explicit 0/1 flags using published clinical thresholds (Table B.2). Flags describe which evidence rules apply; they are not diagnoses.
3. **Layer 3, priorities.** Each evidence rule fires on a flag, sets a base priority (low, medium or high), may raise it when further flags hold, and cites rows of the evidence table (Table B.1). The patient's own 24-hour dietary recall then moves the priority one level up or down. No priority carries a patient-level numeric target.
4. **Layer 4, recipe criteria.** Each priority is linked to one recipe-composition measure with a reference value from public dietary guidance (Table B.5). The priority level sets the weight.
5. **Layer 4, ranking.** Candidate meals are scored for health relevance, cuisine compatibility, practical feasibility and acceptability (each 0–100) and combined with explicit weights (0.6, 0.2, 0.2, 0.1). Recipes that meet every reference value rank first. Each recommendation reports its criterion values, reference values and supporting evidence rows.

6. **Meal plans and Layer 5 scenarios.** Seven-day plans are drawn from the ranked list under weekly constraints, and a scenario compares chosen recipes with the patient's usual diet, giving an evidence-graded direction rather than a predicted value.

B.2 Intervention Evidence Table

The evidence table contains 23 intervention–outcome rows from 11 sources. Every citation was checked against its PubMed record; the check corrected two first authors (Sangouni et al. and Haigh et al.), completed an incomplete citation (Stern et al., 2026) and repaired two guideline rows whose fields had shifted by one column. Null findings are kept so that conflicting evidence is visible. Observational associations (coffee) carry their own label and are recorded but never used as effects.

Table B.1. Intervention evidence rows.

ID	Intervention	Outcome	Strength	Source
VG2015-WL5	Weight loss ≥ 5%	NASH resolution, NAS	Strong	Vilar-Gomez et al., 2015
VG2015-WL7-10	Weight loss 7–10%	NAS	Strong	Vilar-Gomez et al., 2015
VG2015-WL10-NASH	Weight loss ≥ 10%	NASH resolution	Strong	Vilar-Gomez et al., 2015
VG2015-WL10-FIB	Weight loss ≥ 10%	Fibrosis regression	Strong	Vilar-Gomez et al., 2015
SAN2021-AST-GGT	Mediterranean diet	AST, GGT	Moderate	Sangouni et al., 2021
SAN2021-ALT	Mediterranean diet	ALT (no effect)	Weak / inconclusive	Sangouni et al., 2021
STE2026-MEDDIET	Mediterranean diet	ALT, stiffness, MRI-PDFF	Moderate	Stern et al., 2026
HAI2022-DIET	Dietary interventions	ALT, AST, FLI, steatosis	Moderate	Haigh et al., 2022
HAI2022-CR	Calorie restriction	ALT, steatosis, stiffness	Moderate	Haigh et al., 2022
XIO2024-MD-LFD	Mediterranean vs low-fat	Enzymes, liver fat (no difference)	Moderate	Xiong et al., 2024
STI2023-EX	Exercise training	MRI liver fat	Moderate	Stine et al., 2023
STI2023-EXDOSE	Exercise ≥ 750 MET-min/week	MRI liver fat	Moderate	Stine et al., 2023
EASL2024-DIET	Diet quality; limit sugars, saturated fat	Liver injury	Guideline	EASL-EASD-EASO, 2024
EASL2024-PA	Physical activity	Steatosis	Guideline	EASL-EASD-EASO, 2024
LUU2018-SAT	Excess saturated vs unsaturated fat vs sugars	Liver fat	Moderate	Luukkonen et al., 2018
CHI2014-HYPER	Hypercaloric fructose	Liver fat, ALT	Moderate	Chiu et al., 2014
CHI2014-ISO	Isocaloric fructose	Liver fat, ALT (no effect)	Weak / inconclusive	Chiu et al., 2014
LIU2015-COF-FIB	Coffee	Advanced fibrosis, cirrhosis	Observational	Liu et al., 2015
EBA2021-COF-FIB	Coffee	Significant fibrosis in NAFLD	Observational	Ebadi et al., 2021
EBA2021-COF-INC	Coffee	NAFLD incidence (no association)	Observational	Ebadi et al., 2021
STE2026-FAST	Fasting	ALT, liver stiffness	Moderate	Stern et al., 2026

ID	Intervention	Outcome	Strength	Source
STE2026-LCHF	Low-carbohydrate high-fat diet	ALT (no effect)	Weak / inconclusive	Stern et al., 2026
STE2026-OMEGA3	Omega-3 supplementation	ALT (no effect)	Weak / inconclusive	Stern et al., 2026

Strength labels follow the project scheme: STRONG 4, MODERATE 10, WEAK / INCONCLUSIVE 4, guideline recommendation 2, observational association 3. The Luukkonen et al. trial (n = 38) is labelled MODERATE as a controlled trial with a significant effect. Source years follow the study ID.

B.3 Patient State and Priority Rules

Table B.2. Thresholds used to derive patient-state flags.

Flag	Definition	Source
Steatosis likely	P(CAP ≥ 248) ≥ 0.432 (classifier threshold)	Layer 2 (Appendix A)
Moderate–severe steatosis likely	P(CAP ≥ 280) ≥ 0.365 (classifier threshold)	Layer 2 (Appendix A)
Overweight / obesity	BMI ≥ 25 / ≥ 30 kg/m²; ≥ 23 / ≥ 27.5 for Asian participants	WHO; WHO Expert Consultation, 2004
Central obesity	Waist > 102 cm (men), > 88 cm (women)	NCEP ATP III
Elevated glycaemia	HbA1c ≥ 5.7%	American Diabetes Association
Dyslipidaemia	Triglycerides ≥ 150 (fasting) or ≥ 175 mg/dL (non-fasting); HDL < 40 (men) or < 50 mg/dL (women)	NCEP ATP III; EAS/EFLM, 2016
Weight management appropriate	Steatosis likely and (overweight or central obesity)	Project rule

Table B.3. Evidence rules: when each priority applies and how its level is set.

Priority	Applies when	Base level	Raised when	Evidence rows
Weight management	Weight management appropriate	Medium	High if moderate–severe steatosis or obesity	VG2015 (×3), HAI2022-CR
Dietary quality	Steatosis likely	Medium	High if moderate–severe steatosis	EASL2024-DIET, STE2026, HAI2022, SAN2021 (×2), XIO2024
Limit saturated fat	Steatosis likely	Medium	High if dyslipidaemia	EASL2024-DIET, LUU2018-SAT
Limit excess sugar	Steatosis likely	Low	Medium if elevated glycaemia or weight management appropriate	EASL2024-DIET, CHI2014-HYPER, CHI2014-ISO

The sugar priority starts low because feeding trials show harm when fructose adds surplus energy but not when it replaces other carbohydrates (CHI2014-ISO).

Personalisation from the 24-hour recall

NHANES records each participant's intake on the day before the examination. When the recall is reliable, three priorities move one level with the patient's own intake, using the same reference values that score recipes: up when intake misses the reference, down when it meets it. Weight management has no intake measure and is unchanged. One day is a noisy measure

for an individual, so a priority never moves by more than one level and the reason is shown with it.

Table B.4. Recall measures that adjust priorities.

Priority	Recall measure	Reference
Dietary quality	Fibre per 1,000 kcal	≥ 14 g
Limit saturated fat	Saturated fat, % of energy	≤ 10%
Limit excess sugar	Added sugars, % of energy (FPED)	≤ 10%

Recalls are used when coded reliable, the day was usual, and energy is plausible (500–3,500 kcal for women, 800–4,000 for men). FPED added sugars for 2021–2023 are derived from the individual foods with USDA's method, which reproduces USDA's 2017–2020 totals.

Of the 2,040 held-out participants aged 12 to 59, 1,112 (55%) receive at least one priority, almost always all four; participants whose steatosis is unlikely receive none. The recall is what makes the priorities differ between patients:

Table B.5. Personalisation on 300 held-out participants aged 12–59 (same engine, without and with the recall).

Measure	Without recall	With recall
Distinct priority profiles	9	29
Share with the most common profile	43%	16%
Distinct top-10 recipe lists	15	27
Mean top-10 overlap, same cuisine	0.953	0.928

B.4 Recipe Criteria, Scoring and Candidate Screening

Table B.6. Recipe criteria linked to each priority. Scores are 100 at or beyond the reference value and fall linearly to 0 over the tolerance.

Priority	Recipe measure	Reference	Tolerance	Source of reference
Weight management	Energy per serving	≤ 600 kcal	400 kcal	Public Health England 400-600-600, 2018
Limit saturated fat	Saturated fat, % of energy	≤ 10%	10 points	WHO, 2023; DGA 2020–2025
Limit excess sugar	Total sugars, % of energy	≤ 10%	15 points	WHO sugars guideline, 2015
Dietary quality	Fibre per 1,000 kcal	≥ 14 g	14 g	DGA 2020–2025

Reference values describe the composition of a recipe and are the same for every patient; what varies by patient is which criteria apply and how they are weighted (low 1, medium 2, high 3). RecipeDB reports total rather than added sugars, so fruit and milk sugars count towards the sugar criterion.

- **Health gate.** Recipes that meet every reference value rank ahead of those that do not. Inside the gate every health score is 100, so half of the health component is instead the priority-weighted margin past the saturated-fat, sugar and fibre references; a patient with a high saturated-fat priority is therefore offered recipes further below that reference. Energy has no margin, because a much smaller meal is not better for that reason.
- **Acceptability.** The RecipeDB rating shrunk towards the corpus mean (4.91) with a pseudo-count of 20, so five stars from three raters does not beat 4.9 from 3,000; ratings of 4.5–5.0 map to 0–100.
- **Weights.** Health 0.6, cuisine 0.2, feasibility 0.2 and acceptability 0.1, renormalised over the criteria a recipe has. Cuisine counts exact matches and configured related cuisines.

Table B.7. Screening of the RecipeDB corpus before ranking.

Step	Rule	Effect
Nutrition plausibility	Energy > 0 and ≤ 2,500 kcal per serving; sodium ≤ 6,000 mg; macronutrient energy within 30% of stated energy	774 of 20,000 recipes (3.9%) held for review
Main meals only	Excludes non-meal categories (desserts, beverages, sauces, household and beauty items) and titles naming seasonings or condiments; energy ≥ 200 kcal per serving	10,108 candidate meals remain; 9,844 have plausible nutrition
Dietary preference	Every diet tag must be allowed by the requested diet	About 1,000 recipes have conflicting tags (e.g. a prawn dish tagged vegan)

B.5 Evaluation of the Ranking Engine

Following Section 12.2 of the proposal, the ranking was evaluated on 300 randomly selected held-out participants aged 12 to 59 who had at least one priority (of 1,112 eligible), using their 24-hour recall. For each participant, the ten highest-ranked recipes from each method were checked against the three criteria: health (the recipe meets the reference value of every criterion that applies to the patient), cuisine (it matches the requested cuisine) and feasibility (total time ≤ 60 minutes, ≤ 15 ingredients and ≤ 12 steps). NHANES does not record cuisine preferences, so each participant was assigned one at random from the five cuisines with at least 100 candidate meals. Random selection was repeated 20 times per participant.

Table B.8. Share of each top-10 list meeting the criteria (mean over 300 participants aged 12–59).

Method	Health	Cuisine	Feasibility	All three	Diversity
Multi-criteria, health-gated (default)	1.000	0.998	0.832	0.831	0.947
Health-gated, without acceptability	1.000	0.994	0.803	0.796	0.945
Health-gated + diversity re-ranking	1.000	0.988	0.818	0.806	0.956

Method	Health	Cuisine	Feasibility	All three	Diversity
Multi-criteria, weighted only	0.737	1.000	0.970	0.727	0.952
Health score only	1.000	0.057	0.400	0.017	0.966
Nutrition only (non-personalised)	1.000	0.057	0.400	0.017	0.966
Random	0.057	0.053	0.596	0.003	0.962

Diversity: mean pairwise 1 − Jaccard similarity of ingredient sets. Without the recall the default reaches 0.835 on all three criteria.

Only the health-gated variants always meet every evidence-linked reference. Without the gate the weighted score traded health for cuisine or practicality in 26% of recommendations. Acceptability raised the share meeting all three criteria from 0.796 to 0.831 and is part of the default. Diversity re-ranking added little (0.947 to 0.956) at a cost in practicality, because the default list is already varied, so it remains an option. The health-only and nutrition-only baselines coincide because both select among the same tied recipes that meet every reference value.

Priority levels shape the recommendations in the expected direction: with a high saturated-fat priority the top 10 averages 3.25% of energy from saturated fat against 3.55% with a low one, and 4.22% against 4.58% for sugars. Fibre does not differ, because almost every gated recipe is already far above the fibre reference.

The sensitivity analysis reran the default ranking over 36 combinations of weights in steps of 0.1 (Figure B.1). The mean overlap of the top 10 with the default was 0.74 (median 0.78); it fell mainly when cuisine had weight 0.1 (0.50) and was 0.77–0.83 for cuisine weights of 0.2 to 0.7.

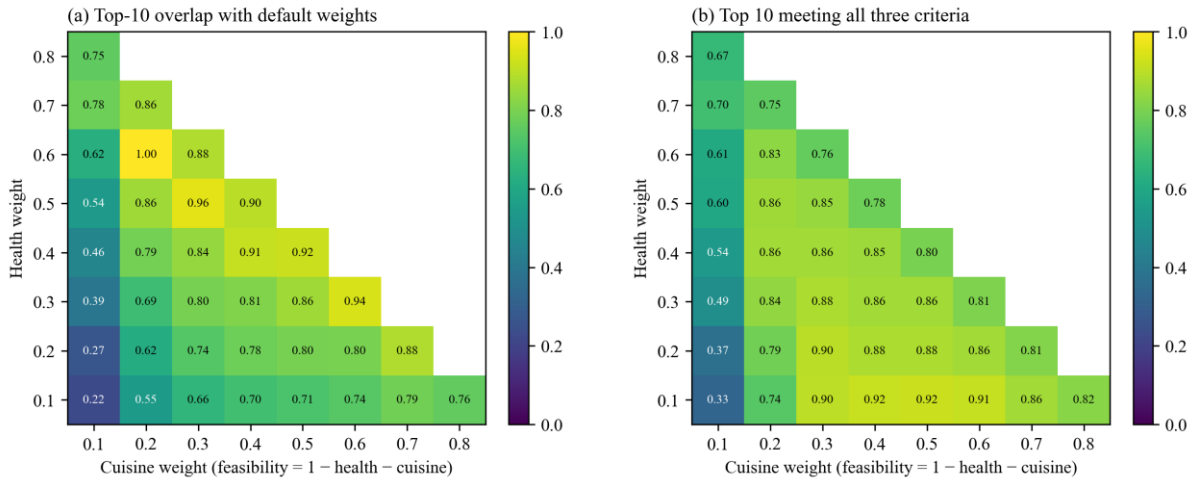


Figure B.1. Weight sensitivity of the default ranking over 300 held-out participants aged 12–59. (a) Mean share of the default top ten that is retained. (b) Mean share of the top ten meeting all three criteria.

B.6 Seven-Day Meal Plans

A plan gives two main meals a day for seven days from recipes that meet every reference, practical recipes first. Each plan is checked for: every meal and every day meeting the references, a recipe at most twice and never on the same or consecutive days, different protein families within a day and at most three per week, and the requested cuisine. Because a patient may lack the ingredients, tools or taste for another cuisine, the week is filled in three steps and stops at the first that completes it: (1) the requested cuisine only; (2) a favourite recipe may appear twice; (3) recipes from the most similar cuisines by ingredient profile (cosine similarity ≥ 0.8), each labelled as borrowed.

Table B.9. Seven-day plans for the 300 participants.

Check	Share of patients
Every meal meets the references	1.000
Every day meets the references	1.000
Repeats within limits; protein varied	1.000
Complete with own cuisine only (step 1)	0.583
Complete by step 2 (a favourite may repeat)	0.803
Complete by step 3 (similar cuisine allowed)	1.000

Asian, European and Mexican plans complete at step 1; American plans at step 2; Canadian plans need step 3 and borrow 0.98 meals on average. Plans used 5.9 protein families; 81% of plan meals are within the practical limits.

B.7 Scenarios (Layer 5)

A scenario is a set of recipes that replace the patient's usual meals. For each recipe criterion the engine compares the scenario's mean composition with the patient's own 24-hour recall (usual meal energy is a third of daily energy) and reports whether it improves, worsens or

does not change meaningfully, with the evidence label of that pathway: energy (calorie restriction, HAI2022-CR, MODERATE; weight loss of 5% or more, VG2015-WL5, STRONG, shown as conditional), saturated fat (LUU2018-SAT, MODERATE), sugars (MODERATE only when energy also falls, otherwise WEAK, following CHI2014-HYPER and CHI2014-ISO) and fibre as a marker of a Mediterranean-style diet (STE2026-MEDDIET, MODERATE). The Layer 2 model is not queried for scenarios, because it was not trained on interventions; every scenario is labelled "Simulated projection, not a clinical forecast".

B.8 Worked Example

Participant 109349 (aged 47, held out from training; measured CAP 338 dB/m, reliable examination) has a predicted CAP of 313.7 dB/m and calibrated probabilities of 0.96 for CAP ≥ 248 and 0.78 for CAP ≥ 280. BMI 40.5 kg/m² and low HDL (45 mg/dL, below 50 for women) set the obesity and dyslipidaemia flags. All four priorities are high:

Table B.10. Priorities for participant 109349 and their reasons.

Priority	Level	Reason shown
Consider evidence-informed weight management	High	Probable moderate-to-severe steatosis (CAP ≥ 280 dB/m)
Consider overall dietary quality	High	Probable moderate-to-severe steatosis (CAP ≥ 280 dB/m); kept: fibre 4.1 g per 1,000 kcal in the 24-hour recall misses the reference ≥ 14
Prefer recipes lower in saturated fat	High	Raised triglycerides or low HDL; kept: saturated fat 11.7% of energy in the 24-hour recall misses the reference ≤ 10
Prefer recipes lower in added sugars	High	Excess weight, where surplus energy from sugar matters most; raised: added sugars 14.9% of energy in the 24-hour recall misses the reference ≤ 10

With Asian cuisine requested, 9,844 candidate meals are scored, 565 meet every reference value and 264 are held for data review.

Table B.11. Top three recommendations for participant 109349 (Asian cuisine requested).

Rank	Recipe	Energy (kcal)	Sat. fat (% energy)	Sugars (% energy)	Fibre (g/1,000 kcal)	Overall
1	Avocado and Prawns in Wasabi	234	8.8	1.4	29.9	83.7
2	Spicy Chickpeas	271	3.0	3.4	31.7	83.5
3	Potatoes With Indian Spices	209	2.2	3.4	25.4	83.0

Table B.12. Scenarios for participant 109349 (European cuisine): usual diet against each scenario.

Scenario	Criterion	Usual	Scenario	Direction	Evidence
Scenario A	Energy per serving (kcal)	631.7	318.9	improves	MODERATE

Scenario	Criterion	Usual	Scenario	Direction	Evidence
Scenario A	Fibre (g per 1,000 kcal)	4.1	39.6	improves	MODERATE
Scenario A	Saturated fat (% of energy)	11.7	1.9	improves	MODERATE
Scenario A	Total sugars (% of energy)	20.8	0.3	improves	MODERATE
Scenario B	Energy per serving (kcal)	631.7	551.7	improves	MODERATE
Scenario B	Fibre (g per 1,000 kcal)	4.1	15.9	improves	MODERATE
Scenario B	Saturated fat (% of energy)	11.7	4.9	improves	MODERATE
Scenario B	Total sugars (% of energy)	20.8	6.5	improves	MODERATE

Scenario A (Broccoli and Garlic Pasta; 5 Ingredient Soup; Southern Lima Butter Beans With Ham Bits): improvement, MODERATE. Scenario B (Mediterranean Quinoa Salad; Pasta Puttanesca; Skillet Penne and Sausage Supper): improvement, MODERATE.

B.9 Limitations

- Reference values come from population dietary guidance, not from liver-specific dose–response evidence; the evidence rows justify the direction of each criterion, not its threshold.
- Most participants with priorities receive all four; personalisation comes from the recall-adjusted priority levels, the margin inside the health gate and the patient's preferences.
- A single 24-hour recall is a noisy measure of usual intake, which is why it moves a priority by one level at most.
- Cuisine preferences were simulated, and 73% of RecipeDB recipes have the cuisine label "Other", which limits cuisine matching. Cuisine compatibility uses exact and configured related matches rather than the ingredient co-occurrence measure described in the proposal.
- RecipeDB nutrition is per serving as reported by recipe authors; implausible values are screened out but plausible errors remain. Total sugars stand in for added sugars in recipes.
- The evaluation covers held-out participants aged 12 to 59, the Layer 2 model's scope.
- The evaluation measures whether recommendations satisfy the defined criteria. It does not show that following them changes liver fat; the planned pilot evaluation of recipe appropriateness has not yet been run.